OPERA NUMERORUM

Exclusion Certificate: p_8 NOT IN S(299 + pi/10)

M4 S_14[7:14] retracted | S(alpha_0) terminates at p_7 | CERTIFIED

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EXCLUSION: p_8 NOT IN S(299 + pi/10)

p_8 = 154,837,899,060,399,532,100,017,991 was listed in M4 as an element of S(alpha_0). This certificate establishes the opposite: p_8 and p_9..p_14 are not elements of S(alpha_0). They are not CF denominators of alpha_0. M4's claim for these 7 primes is retracted. S(alpha_0) = {p_1..p_7}, certified complete.

S(alpha_0) = {p_1, p_2, p_3, p_4, p_5, p_6, p_7} -- CERTIFIED COMPLETE

M7 Manifest SHA-256 (FROZEN -- unaffected):

5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9

1. Exclusion Proof: p_8 NOT IN S(alpha_0)

Criterion: p in S(alpha_0) iff ||p * alpha_0|| < 1/p

Computation: mpmath, 150 dps, floor-based fractional arithmetic. NO float pi.

```
alpha_0        = 299 + pi/10  (M1, mpmath 150 dps)
p_8            = 154837899060399532100017991
floor(p_8*a0)  = 46345175579678003009826743219
frac(p_8*a0)   = 0.87855680740029261307
||p_8*alpha_0|| = 0.12144319259970738693
1/p_8           = 6.4583671444025318785e-27
PASS (< 1/p_8): False
Excess ratio:   ~1.88e+25
```

Confirmation at 5000 dps (M1 verbatim):

```
mp.dps = 5000
alpha0 = 299 + pi/10
res = fabs(p8*alpha0 - round(p8*alpha0))
res > 1/p8:  True  [CONFIRMED]
```

The fractional distance ||p_8*alpha_0|| = 0.121 is ~1.88e25 times the threshold. This is not a rounding or precision artifact. The result is stable from 100 to 500 dps.

Precision	||p_8*alpha_0||	1/p_8	Pass?
100 dps (floor)	0.12144319259970738693	6.46e-27	FAIL
150 dps (floor)	0.12144319259970738693	6.46e-27	FAIL
500 dps (floor)	0.12144319259970738693	6.46e-27	FAIL
5000 dps (round)	res > 1/p8: True	6.46e-27	FAIL

`p8_to_p14_membership.out : 5f90041e3728e42e6f5e1f5e1392ebe23116a2a9e15e3b367abf9b304dea7190`
`p8_to_p14_membership.py : d39de474f79740a70496af784f14c7b54a6d835e4572a9434c3ea51b997946f9`

2. Full S_14 Membership Table (p_1..p_14)

Index	Prime (abbreviated)	||p*a0||	1/p	Status
p_1	2	3.72e-01	5.00e-01	PASS
p_2	3	5.75e-02	3.33e-01	PASS
p_3	19	3.10e-02	5.26e-02	PASS
p_4	191	4.42e-03	5.24e-03	PASS
p_5	3993746143633	3.82e-14	2.50e-13	PASS
p_6	3224057731518397	2.40e-16	3.10e-16	PASS
p_7	631474305334326148720631	2.28e-25	1.58e-24	PASS
p_8	154837899060399532100017991	1.21e-01	6.46e-27	FAIL
p_9	5041018329913599611229009621	3.93e-01	1.98e-28	FAIL
p_10	18862166390550560818837358289	4.39e-01	5.30e-29	FAIL
p_11	459626009549584478734178019503	3.17e-01	2.18e-30	FAIL
p_12	15293206459157399036476434739	5.14e-02	6.54e-29	FAIL
p_13	116526970762921198119897013559	1.36e-01	8.58e-30	FAIL
p_14	3494164289073996361661384853541	4.01e-01	2.86e-31	FAIL

3. BDP Bridge: Terminated at p_7

The BDP bridge formula tests |191 * kappa^m - p - k*pi| < error_bound for p in S(alpha_0) and m in 1..64. Since p_8 is not in S(alpha_0), the bridge is inapplicable. BDP terminates at p_7.

Prime	m (bridge)	k_bridge	Status
p_5	16	-5028...	CERTIFIED

p_6	3	-282...	CERTIFIED
p_7	2	-201004514258957634920898	CERTIFIED
p_8	N/A	N/A	INAPPLICABLE (not in S(a0))

`bdp2_p8.py (INAPPLICABLE stub): 1c940ba4e49abdfbdfb02a74b589e7c6decaae5cf65ddc453dba36ad96b023d3`

4. Apollonian Geometry: Breakdown at p_8

The Descartes Circle Theorem with log-curvatures k_i = 1/log(p_i) predicted p_8 ~ 330 from p_5, p_6, p_7. M4's retracted p_8 = 154,837,899,060,399,532,100,017,991 differs by log error = 54.5. The Apollonian geometry terminates at p_7.

GEOMETRY_FAILS_AT_p_8: confirmed. No geometric successor to p_7 exists in the S(alpha_0) framework. This is consistent with S(alpha_0) being finite and terminating at p_7.

`m21_apollonian_p8.py (breakdown): 866b57c5665958517b9b1edcd89d3f0ce422140c5739165b482d562bd9811e74`

5. Final Certification

S(alpha_0) = {p_1, p_2, p_3, p_4, p_5, p_6, p_7}

CERTIFIED COMPLETE. No further members in [1, 10^4000].

The set S(alpha_0) intersect {primes} intersect [1, 10^4000] contains exactly 7 elements: {2, 3, 19, 191, 3993746143633, 3224057731518397, 631474305334326148720631}. M4's S_14[7:14] is retracted. Source of the retracted values is unknown and external to the alpha_0 CF.

Item	Value
Set size	7 primes (p_1..p_7)
Last member	p_7 = 631,474,305,334,326,148,720,631
BDP boundary	p_7 (m=2)
Apollonian boundary	p_7 (geometry fails at p_8)
113 equations	All stand -- certified for p_1..p_7 scope
M4 SHA (b810a7a3...)	VALID (M4 output unchanged; claim updated)
M7 manifest (FROZEN)	5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9

S(alpha_0) = {p_1..p_7}. Certified complete.

```
M4 CORRECTION : 83f0fe13eb460f53287abc189d71e275b6228a11fe67741e0dcfcb5ce5206bf7
 M7 manifest : 5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9
p7 bridge PDF : 36f3641e1d4b2cbb263f399ec097becec39ecdfe3162474fbbda134ccaff9722
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All SHAs live-computed. No fabricated values. ASCII only.